Introduction to Python

BEGINNER LESSON | Qspiders | Python 3

Learning goals: Explain what Python is, recognize its common uses, run a simple program, and understand basic syntax.

1. What is Python?

Python is a general-purpose programming language. A program is a set of instructions that tells a computer what to do. Python uses readable syntax, which makes it approachable for beginners.

Guido van Rossum created Python, and its first public release was in 1991. The name was inspired by the comedy group Monty Python.

2. Where is Python used?

• Automation: rename files, process spreadsheets, and reduce repetitive work.
• Web development: build the server-side logic of websites.
• Data analysis: clean, explore, and summarize data.
• Artificial intelligence: build and use machine-learning systems.
• Testing and scripting: check software and connect tools.

3. Key characteristics

Readable: Code is designed to be easy to follow.
High-level: Many low-level details are handled for you.
Dynamically typed: You do not have to declare a variable's type before assigning a value.
Cross-platform: Python runs on major operating systems.
Rich ecosystem: The standard library and third-party packages provide reusable tools.

Python is commonly described as interpreted. In CPython, the usual implementation, source code is compiled to bytecode and then executed by the Python virtual machine.

Your First Python Program

4. Ways to run Python

Install Python 3 from the official Python website if needed. You can type instructions in an interactive shell, or save them in a file ending in **.py** and run that file. An editor or IDE helps you write and run programs.

In a terminal, check whether Python is available. On Windows, the Python launcher may provide the py command:

```
py --version
# Alternatively, if available:
python --version
```

5. Hello, world!

Create a file named **hello.py** and write:

```
print("Hello, world!")
```

Run it from the folder containing the file:

```
py hello.py
# Or: python hello.py
```

Output: Hello, world!

The **print()** function displays a value. Quotation marks identify text, called a string. Parentheses contain the value passed to the function.

6. Comments and simple calculations

```
# This is a comment: Python ignores it.
print("Welcome to Python")
print(2 + 3)
print(10 * 2)
```

The last two lines display **5** and **20**. Text needs quotation marks; arithmetic expressions do not.

Syntax and Practice

7. Rules to remember

Case matters: print and Print are different names.
Indentation matters: Indented lines form a block. Use four spaces consistently.
Quotes must match: Both single and double quotes can delimit strings.
One instruction per line: This is a good beginner habit.

```
if 5 > 2:
    print("Five is greater than two")
```

The colon starts the block, and the indented line belongs to the condition. You will study conditions in more detail later.

8. Common beginner errors

SyntaxError: Python cannot understand the structure, such as a missing closing quote or parenthesis.
NameError: A name has not been defined, such as typing Print instead of print.
IndentationError: The indentation is missing or inconsistent.

Read the final line of the error message, check the indicated line, correct it, and run the program again.

9. Try it yourself

1. Display your name and a welcome message on separate lines.
2. Display the result of 12 + 8.
3. Add a comment describing what your program does.
4. Explain the difference between print("2 + 3") and print(2 + 3).

Suggested answers

```
# My first practice program
print("My name is Alex")
print("Welcome to my Python lesson")
print(12 + 8)
```

For question 4, the first instruction displays the text **2 + 3**; the second calculates the expression and displays **5**.

Next lessons: Variables, data types, and operators.

Further reading: Official Python tutorial | Python downloads